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Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 6: Technical Safety Concept (TSC)

Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	Technical Safety Concept (TSC).....	6
4.1	Dual-Path Validation (CPU and Safing Engine).....	6
4.2	Watchdog Supervision	6
4.3	Safe State Enforcement.....	7
4.4	Output Driver Hardware	7
4.5	Power Supply Safety.....	7
4.6	CAN Communication Safety.....	8
4.7	Hardware Requirements Traceability	9
5	Figure and table listing.....	10
6	List of dependent documents	11
7	Literature	12
8	Version history	13

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	10:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Technical Safety Concept (TSC)

The Technical Safety Concept (TSC) translates Functional Safety Requirements into hardware diagnostics. To satisfy the ASIL D mandate, the following safety features and mechanisms are implemented.

4.1 Dual-Path Validation (CPU and Safing Engine)

Deployment is allowed only when both conditions are satisfied: Main CPU confirms the crash event AND Safing Engine asserts the safing signal. In a production Airbag ECU, High-side and low-side MOSFET switches in series (safing switch architecture) both must be independently commanded HIGH before current can flow through the squib. Fire Enable signal from the MCU must be accompanied by the hardware safing enable for deployment to occur (two-point safety architecture).

Tabelle 2 Safety Feature

Safety Feature	Reference
Dual-path validation (CPU and Safing Engine)	Airbag_ECU_Phase_2, 4.1 Identified Hazards, Table 2, H1, H2, H3
Redundant sensors (electronic and mechanical)	Airbag_ECU_Phase_2, 4.1 Identified Hazards, Table 2, H1, H2
Watchdog supervision	Airbag_ECU_Phase_3, 4.2 Technical Safety Mechanisms, Table 3
ECC/CRC memory protection	Airbag_ECU_Phase_2, 4.1 Identified Hazards, Table 2, H1-H5
Backup energy source for deployment feasibility	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG2
Hardware-based safing signal gating	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG2, SG4

4.2 Watchdog Supervision

A hardware watchdog timer must be implemented to detect MCU software stalls or infinite loops. If the software fails to reset the watchdog within the defined timeout window (typically 10-100 ms), the watchdog hardware resets the MCU. On reset, all output pins default to LOW this is the inherent safe state of the digital outputs.

Fault handling: Watchdog reset on CPU failure, Fault logging in non-volatile memory, Diagnostic message via CAN, Safe state enforcement (no unintended deployment).

4.3 Safe State Enforcement

The hardware safe state ensures that airbag deployment cannot occur accidentally when a hardware fault is detected. The safe state is defined as: Driver airbag output (D8) = LOW, Passenger airbag output (D9) = LOW, Warning lamp (D13) = HIGH, CAN fault frame transmitted with fault code in Byte 6.

Tabelle 3 Safe State Trigger Conditions

ID	Trigger Condition	Hardware Implication
FS01	ADC value out of valid range (less than 2 or greater than 1020)	Sensor open/short circuit; hardware fault — block deployment
FS02	Output readback mismatch (commanded HIGH, reads LOW)	Output driver failure; hardware fault — activate warning lamp
FS03	CAN bus error detected (bus-off state)	Communication hardware fault — enter safe state
FS04	Baby seat signal intermediate voltage (neither 0 V nor 3.3 V)	Seat switch hardware fault — passenger airbag suppressed, warn
FS05	Watchdog timer expiry (MCU stall)	Hardware watchdog resets MCU; outputs go to default LOW (safe)

4.4 Output Driver Hardware

Tabelle 4 Output Driver Hardware

Output Function	Pin	Logic Active	Description
Driver Airbag Deployment	D8	HIGH (3.3 V)	Activates when valid crash + safing condition TRUE
Passenger Airbag Deployment	D9	HIGH (3.3 V)	Activates only when crash valid AND baby seat NOT detected
Warning Lamp (Fault Indicator)	D13	HIGH (3.3 V)	Illuminated during safe state or detected fault condition

4.5 Power Supply Safety

Tabelle 5 Power Supply Safety Architecture

Supply Stage	Specification
Input Voltage Range	9-16 V DC (vehicle battery nominal 12 V)
Primary Regulator	5 V LDO or switching DC-DC converter for MCU supply
MCU Core Supply	3.3 V (Arduino Due operates at 3.3 V logic)
Post-Crash Hold-Up	Capacitor bank (≥ 150 ms at full load) — powers squib firing and MCU after battery loss
Reverse Polarity Protection	P-channel MOSFET or Schottky diode at battery input
Overvoltage Protection	TVS diode at input to clamp transients (load dump: up to 40 V per ISO 7637-2)

4.6 CAN Communication Safety

Tabelle 6 CAN Bus Parameters

CAN Parameter	Value
CAN Standard	ISO 11898 / CAN 2.0B
Message ID	0x120 (high priority airbag frame)
Data Length Code (DLC)	8 bytes
Typical Bus Speed	500 kbit/s (automotive standard)
Termination	120 Ohm at each end of the CAN bus

Tabelle 7 CAN Frame Structure

Byte	Field Name	Content / Meaning
Byte 0	System Status	0x00 = Normal, 0x01 = Crash detected, 0xFF = Fault
Byte 1	Acceleration Value	Raw ADC reading from A0 (0-255 scaled)
Byte 2	Gyroscope Value	Raw ADC reading from A1 (0-255 scaled)
Byte 3	Pressure Value	Raw ADC reading from A2 (0-255 scaled)
Byte 4	Seat Status	0x00 = No baby seat, 0x01 = Baby seat detected
Byte 5	Deployment Status	Bit 0 = Driver airbag, Bit 1 = Passenger airbag
Byte 6	Fault Code	0x00 = No fault, 0x01 to 0xFF = Specific fault codes
Byte 7	Checksum	XOR checksum of Bytes 0-6 for frame integrity

4.7 Hardware Requirements Traceability

Tabelle 8 Hardware Requirements Traceability

SWR ID	Software Requirement	Supporting Hardware Design Element
SWR01	Read acceleration sensor input	Analog input A0, 10-bit ADC, potentiometer simulation (production: MEMS accelerometer)
SWR02	Read gyroscope angular sensor input	Analog input A1, 10-bit ADC (production: MEMS gyroscope with signal conditioning)
SWR03	Read pressure sensor input	Analog input A2, 10-bit ADC (production: MEMS pressure sensor + low-pass filter)
SWR04	Read passenger seat digital input	Digital input D2 with INPUT_PULLUP; active-LOW logic
SWR05	Activate driver airbag output	Digital output D8 (production: high/low-side MOSFET squib driver)
SWR06	Activate passenger airbag (no baby seat)	Digital output D9 with D2 seat switch interlock logic
SWR07	Activate warning lamp during fault	Digital output D13 (active HIGH)
SWR08	Send crash/fault status via CAN	Arduino Due CAN0 controller + MCP2551 transceiver, ID 0x120
SWR09	Read CAN messages for diagnostic/reset	CAN0 receive interrupt handler; due_can library
SWR10	Read back output status after activation	GPIO readback of D8/D9 after asserting; hardware feedback path in production
SWR11	Enter safe state if fault detected	Hardware watchdog, output default-LOW on reset, warning lamp D13
SWR12	Support test routines for module/integration testing	Serial/USB debug interface; Wokwi simulation platform for hardware-in-loop testing

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 Safety Feature	6
Tabelle 3 Safe State Trigger Conditions	7
Tabelle 4 Output Driver Hardware	7
Tabelle 5 Power Supply Safety Architecture.....	7
Tabelle 6 CAN Bus Parameters.....	8
Tabelle 7 CAN Frame Structure	8
Tabelle 8 Hardware Requirements Traceability.....	9

No table of figures entries found.

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
3	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf
4	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx

7 Literature

No. Reference

- [1] International Organization for Standardization (ISO), ISO 26262: Functional Safety Standard for Modern Road Vehicles. https://fscdn.rohm.com/en/products/databook/white_paper/iso26262_wpe.pdf
- [2] S. N. Easwaran, M. Mollat, D. Sreedharan et al., Fault Tolerant Low Side Driver Circuit Design Using Design FMEA for Automotive Applications, IEEE NEWCAS, 2019. doi: 10.1109/newcas44328.2019.8961225
- [3] G. Bahig and A. El-Kadi, Formal Verification of Automotive Design in Compliance With ISO 26262, IEEE Access, vol. 5, pp. 4505–4516, 2017. doi: 10.1109/access.2017.2683508
- [4] M. R. Kabir, S. Boddupalli, A. P. D. Nath, and Sandip Ray, Automotive Functional Safety: Scope, Standards, and Perspectives on Practice, IEEE Consumer Electronics Magazine, vol. 14, no. 1, pp. 10–25, 2025. doi: 10.1109/mce.2024.3394165
- [5] S. Noh, Enhancing AUTOSAR safety mechanisms for ISO 26262 functional safety requirements, International Journal of Automotive Technology, vol. 17, no. 4, pp. 715–723, 2016.
- [6] M. U. Ali et al., Intelligent Driver Drowsiness Detection for Road Safety.
- [7] U. Jain, Software-in-the-Loop (SiL) Simulation and Validation of Airbag ECU, I4Tech, 2025. doi: 10.1109/i4tech64670.2025.11277665
- [8] R. Mariani, RISC-V Functional Safety for Autonomous Automotive Systems, arXiv:2604.17391, 2026. doi: 10.48550/arXiv.2604.17391
- [9] P. Kilian, O. Koller, P. Van Bergen et al., Emergency Operation in the Power Supply Domain According to ISO 26262, IEEE Access, vol. 10, pp. 47557–47569, 2022. doi: 10.1109/ACCESS.2022.3170903
- [10] Lee, K.J., Ki, Y.H., Cheon, J.S. et al., Approach to functional safety-compliant ECU design for electro-mechanical brake systems, Int. J. Automot. Technol., vol. 15, pp. 325–332, 2014. doi: 10.1007/s12239-014-0033-7
- [12] Y. Li, Q. Liu, and C. Cao, Integrated Safety Validation of Connected Automated Vehicles Using Distributed Hardware Redundancy Architectures, Security and Safety, vol. 2, p. 2022009, 2023. doi: 10.1051/sands/2022009

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_6	09.06.2026	v1.00 Initial version. Technical Safety Concept compiled from HW Concept Design Report V02 Sections 6.3, 7, 8, 9, 10 and Architecture V2 Sections 5, 7.